

Human AI Workflow

Project: Verified Study Buddy

1. Human + AI Workflow

Teach the learning cycle

Primary target: Explain the mechanism and terminology of human review and consequential use

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Safety Boundaries

Without opening the old lesson, explain one key idea from Safety Boundaries. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of untrusted input boundaries

Reference example, not the only acceptable wording: Teacher explanation: Untrusted input must be treated according to a contract before reaching code, queries or tools. Data that resembles instructions does not acquire execution authority. For the toy lookup accept room IDs matching one uppercase letter plus one digit. Perform exact dictionary lookup; never evaluate user strings as code. Reject invalid length or characters.

Copyright and Provenance

Without opening the old lesson, explain one key idea from Copyright and Provenance. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of copyright and permitted sources

Reference example, not the only acceptable wording: Teacher explanation: Permission to access a work and permission to redistribute or modify it are separate. Use a classroom licence ledger instead of assuming all online material is reusable. For this fictional policy, licence L permits classroom display with attribution but forbids public redistribution. Licence M permits adaptation and redistribution with attribution. Unknown permission means seek approval or choose a permitted alternative; this exercise is not a legal ruling.

Evaluation Sets

Without opening the old lesson, explain one key idea from Evaluation Sets. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the mechanism and terminology of system evaluation sets

Reference example, not the only acceptable wording: Teacher explanation: A system evaluation set contains cases with expected behaviour and scoring rules. It should cover retrieval, support, fallback and permissions rather than only easy answerable questions. Record case ID, evidence version, expected claims or fallback, permitted actions and scoring rubric. Separate development from final cases. Aggregate metrics only after checking required conditions individually.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A student asks a helper to submit an unchecked generated essay under their name. Design a workflow that preserves student reasoning and follows classroom rules.

Concept to explain

Allocate tasks according to strengths, accountability, and verification needs.

Term · Meaning

Term: workflow

Meaning: An organised sequence of tasks, reviews and handoffs used to complete work.

Term: accountability

Meaning: Responsibility for explaining decisions, checking outcomes and addressing mistakes.

Term: verification

Meaning: Checking a claim or result against an appropriate independent source, calculation or observation.

Term: sign-off

Meaning: Explicit recorded approval that specified work meets the required review conditions.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Two 45-minute sessions. Human review needs a responsible actor, evidence packet and explicit authority. Adding a person to a workflow does not guarantee effective oversight. Use the supplied constructed scenarios to make the mechanism inspectable. Collect W4 before sharing solutions. Students should name intermediate evidence and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the inputs, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from predictions or unmeasured conditions. If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

End goal for the pictorial case

By the end, I can explain human ai workflow using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

1. AI drafts questions from approved notes.
2. Learner checks clarity; retrieval evidence supports claims.
3. Teacher decides whether material is appropriate.
4. Product records who verified and approved without shifting accountability to AI.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

An AI drafts, verifies and submits a student's assessment without review.

Where should responsibility remain?

Exact explanation to model

The student reviews, verifies, follows class rules and remains responsible. Submission requires authorisation and must not bypass learning.

Explain the added worked example

Human review needs a responsible actor, evidence packet and explicit authority. Adding a person to a workflow does not guarantee effective oversight.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

For this fictional workflow the assistant drafts a room request but only the coordinator can approve and submit it. Review packets include sources, uncertainty, requested action and the exact version. Any material change requires new approval.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term · Meaning

Term: Review packet

Meaning: Evidence and proposed action supplied to a reviewer

Term: Version-bound approval

Meaning: Permission tied to an exact proposal version

Term: Escalation

Meaning: Routing a decision to an authorised responsible person

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Coordinator approves draft v1 for room B. Assistant changes room to C in v2 and submits using the old approval.

The submission violates version-bound approval. Preserve v1 approval, request review of v2 and prevent submission until its exact action is approved.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Human AI Workflow

A human approval belongs to the exact draft and action reviewed.

The facts we will use

A teacher approves study handout H1 for classroom printing. H2 adds a pupil email address and proposes public posting. No H2 approval exists.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Human AI Workflow

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify approval

The recorded approval covers H1 and classroom printing.



2. Inspect change

H2 changes both content and distribution scope.



3. Require review

Stop public posting and show H2's changed content and intended audience to the reviewer.

CHANGE ONE THING

H2 is replaced with unchanged H1 for classroom printing.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Identify approval

What happens and why: The recorded approval covers H1 and classroom printing.

Step: 2. Inspect change

What happens and why: H2 changes both content and distribution scope.

Step: 3. Require review

What happens and why: Stop public posting and show H2's changed content and intended audience to the reviewer.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Stop public posting and show H2's changed content and intended audience to the reviewer. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Human review helps only when the reviewer sees the material changes, evidence and actual consequences.

Change one thing in the pictured case

Change: H2 is replaced with unchanged H1 for classroom printing.

Prediction: The original approval covers that declared action.

Reason: Both the approved content and its distribution scope now match.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: An AI drafts, verifies and submits a student's assessment without review.

Write the question to explain: Where should responsibility remain?

Reveal after predictions: The student reviews, verifies, follows class rules and remains responsible.

Submission requires authorisation and must not bypass learning.

Keep for an independent attempt: A workflow drafts a class email, checks its claims and then sends it.

Where should approval occur, and what should the approver see?

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: The submission violates version-bound approval. Preserve v1 approval, request review of v2 and prevent submission until its exact action is approved. Qualification: Review effectiveness depends on usable evidence, time and authority; a checkbox is not evidence of informed review. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the mechanism; reject a claim resting on unmeasured evidence.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1: 0–7 minutes

Teacher action: Retrieve the prerequisite; identify the input, procedure and evidence source.

Stage: 7–18 minutes

Teacher action: Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.

Stage: 18–32 minutes

Teacher action: Trace the declared procedure and worked case. Complete W2, recording intermediate steps.

Stage: 32–45 minutes

Teacher action: Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.

Stage: Session 2: 0–8 minutes

Teacher action: Retrieve the worked reasoning and collect the guided response before discussing it.

Stage: 8–20 minutes

Teacher action: Compare a proposed boundary variant with the original case; identify what remains fixed.

Stage: 20–35 minutes

Teacher action: Collect W4 independently. Require the procedure trace and the bounded conclusion.

Stage: 35–45 minutes

Teacher action: Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Completion task: Coordinator approves draft v1 for room B. Assistant changes room to C in v2 and submits using the old approval. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: The submission violates version-bound approval. Preserve v1 approval, request review of v2 and prevent submission until its exact action is approved.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Human review and consequential use. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Review effectiveness depends on usable evidence, time and authority; a checkbox is not evidence of informed review.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Why can H1's approval not automatically approve H2? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

The reviewer did not approve the added personal detail or public audience. Approval must match the version and action. Changed-case check: The original approval covers that declared action. Both the approved content and its distribution scope now match.

Changed-case reasoning: The original approval covers that declared action. Both the approved content and its distribution scope now match.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Allocate tasks on a workflow map, justify each owner, simulate a failed tool or unsupported claim, and verify the stop/escalation path.

What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The worked result establishes every future case: Review effectiveness depends on usable evidence, time and authority; a checkbox is not evidence of informed review.

A missing input may be replaced without recording the change: A replacement changes the evidence or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent evidence: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

The student reviews, verifies, follows class rules and remains responsible. Submission requires authorisation and must not bypass learning.

Independent question

A workflow drafts a class email, checks its claims and then sends it. Where should approval occur, and what should the approver see?

Expected independent answer

Approval must occur before sending. The person should see the exact recipients, subject, message and relevant verification results.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Approval of a general idea also approves any later message and recipients.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Student plans and drafts, uses permitted assistance, verifies claims and sources, revises in their own understanding and discloses assistance as required by class rules. Submission remains an authorised student action; the tool must not bypass review or the learning task.

Explanations for the additional workbook examples

W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Teacher explanation: Human review needs a responsible actor, evidence packet and explicit authority. Adding a person to a workflow does not guarantee effective oversight. For this fictional workflow the assistant drafts a room request but only the coordinator can approve and submit it. Review packets include sources, uncertainty, requested action and the exact version. Any material change requires new approval.

W2 Trace the worked case

Coordinator approves draft v1 for room B. Assistant changes room to C in v2 and submits using the old approval. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: The submission violates version-bound approval. Preserve v1 approval, request review of v2 and prevent submission until its exact action is approved.

W3 Audit the limit

A report omits this qualification: Review effectiveness depends on usable evidence, time and authority; a checkbox is not evidence of informed review. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: Review effectiveness depends on usable evidence, time and authority; a checkbox is not evidence of informed review. State the observed result separately from the unverified or out-of-scope claim; preserve the original evidence rather than inventing a missing fact.

W4 Apply a new case independently

Teacher approves read-only report R1. System adds email distribution in R2. What permission is missing and what should the reviewer see?

Teacher explanation: Email distribution is a new side effect not covered by report approval. Show recipient, content/version and purpose; require authorised approval before sending.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked input or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure. Require preserved settings, original outputs and data roles. Review effectiveness depends on usable evidence, time and authority; a checkbox is not evidence of informed review. The new case should test this particular mechanism rather than merely restate the original result.

Assess independent transfer

Selected case: Teacher approves read-only report R1. System adds email distribution in R2. What permission is missing and what should the reviewer see?

Selected case answer

Teacher explanation: Email distribution is a new side effect not covered by report approval. Show recipient, content/version and purpose; require authorised approval before sending.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Verified Study Buddy

AI drafts; sources ground; validators check; learner chooses; teacher governs classroom use.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record.

Retrieve this record next week rather than restarting the project.

Support and stretch

Provide one short approved passage; separate instruction, input, output and evidence. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 33: Capstone Design. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Provide an input/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace. Extension: compare two concrete variants of this mechanism while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Email distribution is a new side effect not covered by report approval. Show recipient, content/version and purpose; require authorised approval before sending. Require intermediate reasoning and the stated scope. Fresh retry: Approval permits sending to one named recipient; proposed action adds two more. Expected reasoning: Recipient scope changed; block the expanded send and request review of the new destination list. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Next connection: Latency cost and resource use. Preserve the current case record and retrieve its evidence boundary before moving on.

Plan the delayed return

Suggested return points in this level: ai-3/33, ai-3/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

7. Prepare, teach and assess

Teaching target: Explain the mechanism and terminology of human review and consequential use

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Coordinator approves draft v1 for room B. Assistant changes room to C in v2 and submits using the old approval. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: The submission violates version-bound approval. Preserve v1 approval, request review of v2 and prevent submission until its exact action is approved.

Check every learner before discussion: Use the worked case for Human review and consequential use. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Review effectiveness depends on usable evidence, time and authority; a checkbox is not evidence of informed review.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: H2 is replaced with unchanged H1 for classroom printing. Require a prediction and a reason before comparing. Expected result: The original approval covers that declared action. Reason: Both the approved content and its distribution scope now match.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.